Method of Gauss-based Orbit Determination: Locating and Identifying the 1950 LA Asteroid[5]

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Introduction

Object of Study: Our study follows the Near-Earth Asteroid **1950 LA**. NEA candidates are typically classified as asteroids in space with a **perihelion distance of less than 1.3 AU.** Our asteroid had a consistent altitude of over 30 degrees to ensure consistent potential for observations.

Asteroids, often referred to as “minor planets," date back to the beginnings of the solar system, where excess rocky material coalesced to form clumps known as planetesimals, eventually becoming planets.

The Six Keplerian Orbital Elements provide a complete description of an asteroid's orbit by defining its size, shape, orientation, and position at a given time.

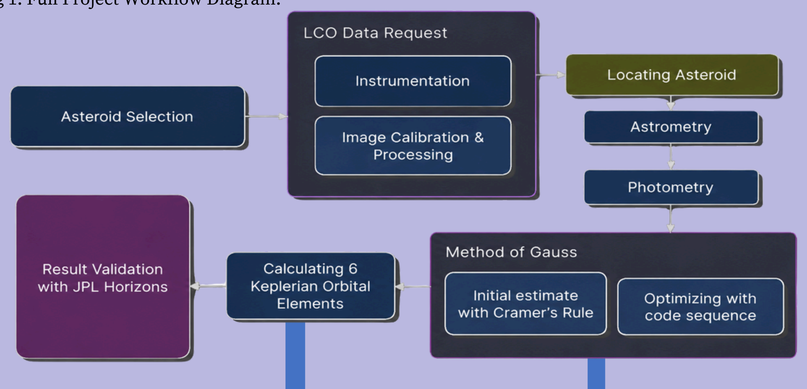
- The **semimajor axis** determines the overall size of the orbit, the **eccentricity** describes its shape, and the **inclination** specifies the tilt of the orbital plane relative to the ecliptic. The **longitude of the ascending node** and the **argument of perihelion** define the orientation of the orbit in three-dimensional space, while the **mean anomaly** indicates the asteroid's position along its orbit at the chosen epoch.

Research Objectives

1. Determine how classical celestial mechanics enables the accurate determination of **asteroid trajectory**
2. Using three observations of the right ascension and declination, determine an asteroid’s trajectory using the **six orbital elements**

Methods

Fig 1. Full Project Workflow Diagram.



Keplerian orbital elements:

- Define an asteroid orbit’s size, shape, orientation, and location with the six parameters: a, e, i, Ω, ω, M.

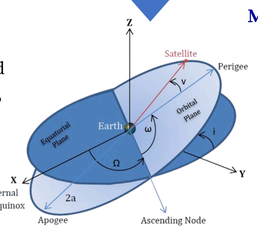


Fig 2. Illustration of 6 orbital elements.

Method of Gauss:

- Estimates an asteroid's position and velocity vectors from three RA & Dec optical observations

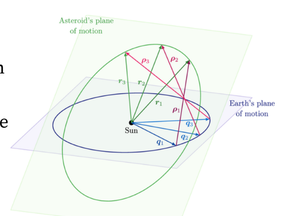


Fig 3. Method of Gauss Visualization.

Focus #1: Target Identification

Asteroid Selection:

The **1950 LA** asteroid has a semimajor axis of 1.709 AU, orbital eccentricity of 0.365, and a perihelion distance of 1.0857 AU.

LCO Data Acquisition:

Asteroid images were requested from the Las Cumbres Observatory network.

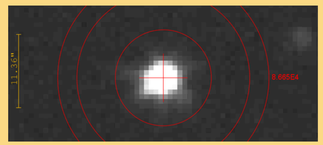


Fig 4. 1950 LA Asteroid on AstroImageJ.

1 **Open two image frames of the asteroid in AstroImageJ**

2 **Blink between the two frames to visually detect asteroid movement**

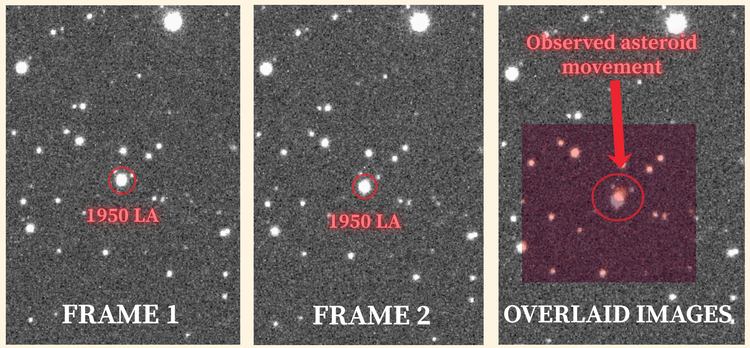


Fig 5-7. 1950 LA movement on June 20th 6:15-6:20 AM UTC.

3 **Note the stationary nature of background reference stars in contrast and record asteroid RA and DEC through centroiding**

Focus #2: Monte Carlo Simulation

Determined how **small perturbations in inputs (RA and Dec) influenced the outputs**

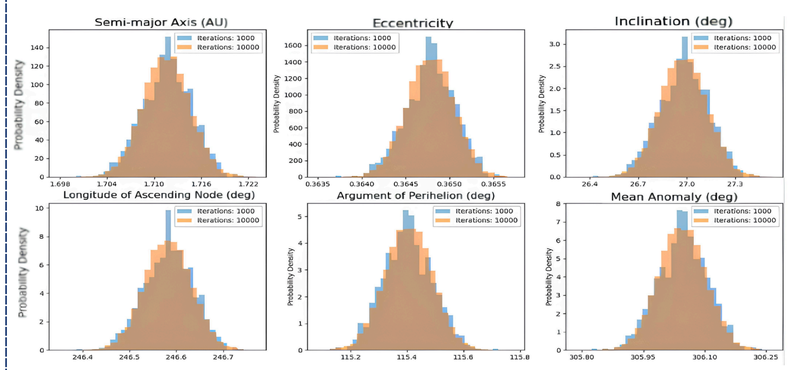


Fig 8. Monte Carlo Simulation of the Six Orbital Elements with **n=1000** and **n=10000** iterations.

- When the number of iterations was increased, smoothness of the estimated distributions improved, producing consistent orbital elements statistics.
- **These results indicate that our solution is robust to measurement uncertainty.**

Results: Method of Gauss

Orbital Element	Calculated Value	Percent Error
Semimajor axis, a (AU)	1.712	0.128%
Eccentricity, e	0.365	0.022%
Inclination, i (°)	26.978	0.402%
Longitude of ascending node, Ω (°)	246.582	0.004%
Argument of perihelion, ω (°)	115.407	0.052%
Mean anomaly, M (°)	306.044	0.007%

Table 1. MoG and Orbital Elements results and error table.

The following table shows a **summary** of the **results** obtained from our workflow. These results indicate good numerical **convergence** and **very accurate** orbital determinations from three nights of observations.

Conclusion and Discussion

Results

- The Method of Gauss was able to accurately recover all six orbital elements for 1950 LA, taking 42 iterations before convergence occurred at a level of optimization error of 3.201×10^{-15}.
- All orbital element calculations were within ± 0.5% agreement with JPL Horizons.

Limitations and Improvements

- The use of only a three day observation arc resulted in both an unstable and non-physical solution.
- A twenty-three day arc, which increased the angular separation, gave a physically reasonable and accurate orbit, yet could still be lengthened for more accuracy.
- There are still some remaining sources of uncertainty associated with this orbit calculation. These include uncorrected light time effects, gravitational perturbations from other bodies in our solar system, and atmospheric refraction.

Conclusion

- The results obtained demonstrate that the calculated orbital elements for Asteroid 1950 LA were extremely similar to those found by JPL Horizons.
- These results highlight the importance in better orbit determination not only for planetary defense, but to explore and better understand our Solar System.

Selected References

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